

HACK THE BOX SHERLOCK - DFIR CROWNJEWEL-2

Different from previous methods, where I used Kali Linux and the tools are Chainsaw, strings and grep, and jq. This time I setup a Windows virtual machine, and installed some of Eric Zimmerman tools, including Timeline Explorer (to view the CSV) and EvtxECmd (to convert the evtx files into csv format

```
C:\Users\Rafael.WINDOWS10\Desktop\HacktheBox Sherlock\CrownJewel2\Artifacts>EvtxECmd.exe -f SYSTEM.evtx --csv CSV --csvf system.csv
EvtxECmd version 1.5.2.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/evtx

Command line: -f SYSTEM.evtx --csv CSV --csvf system.csv

Warning: Administrator privileges not found!

CSV output will be saved to CSV\system.csv

Maps loaded: 453

Processing C:\Users\Rafael.WINDOWS10\Desktop\HacktheBox Sherlock\CrownJewel2\Artifacts\SYSTEM.evtx...
Chunk count: 2, Iterating records...
Record #: 4 (timestamp: 2024-05-15 05:35:07.5120124): Warning! Time just went backwards! Last seen time before change: 2024-05-15 05:35:18.1034501

Event log details
Flags: None
Chunk count: 2
Stored/Calculated CRC: E200F792/E200F792
Earliest timestamp: 2024-05-15 05:35:07.5120124
Latest timestamp: 2024-05-15 05:42:18.0809048
Total event log records found: 208

Records included: 208 Errors: 0 Events dropped: 0
```

```
C:\Users\Rafael.WINDOWS10\Desktop\HacktheBox Sherlock\CrownJewel2\Artifacts>EvtxECmd.exe -f SECURITY.evtx --csv CSV --csvf security.csv
EvtxECmd version 1.5.2.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/evtx

Command line: -f SECURITY.evtx --csv CSV --csvf security.csv

Warning: Administrator privileges not found!

CSV output will be saved to CSV\security.csv

Maps loaded: 453

Processing C:\Users\Rafael.WINDOWS10\Desktop\HacktheBox Sherlock\CrownJewel2\Artifacts\SECURITY.evtx...
Chunk count: 4, Iterating records...

Event log details
Flags: None
Chunk count: 4
Stored/Calculated CRC: 1192ACF3/1192ACF3
Earliest timestamp: 2024-05-15 05:35:11.7435880
Latest timestamp: 2024-05-15 05:41:18.7224829
Total event log records found: 267

Records included: 267 Errors: 0 Events dropped: 0
```

Rafael Angelo Christianto

```
C:\Users\Rafael.WINDOWS10\Desktop\HacktheBox Sherlock\CrownJewel2\Artifacts>EvtxECmd.exe -f APPLICATION.evtx --csv CSV
-csvf application.csv
EvtxECmd version 1.5.2.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/evt

Command line: -f APPLICATION.evtx --csv CSV --csvf application.csv

Warning: Administrator privileges not found!

CSV output will be saved to CSV\application.csv

Maps loaded: 453

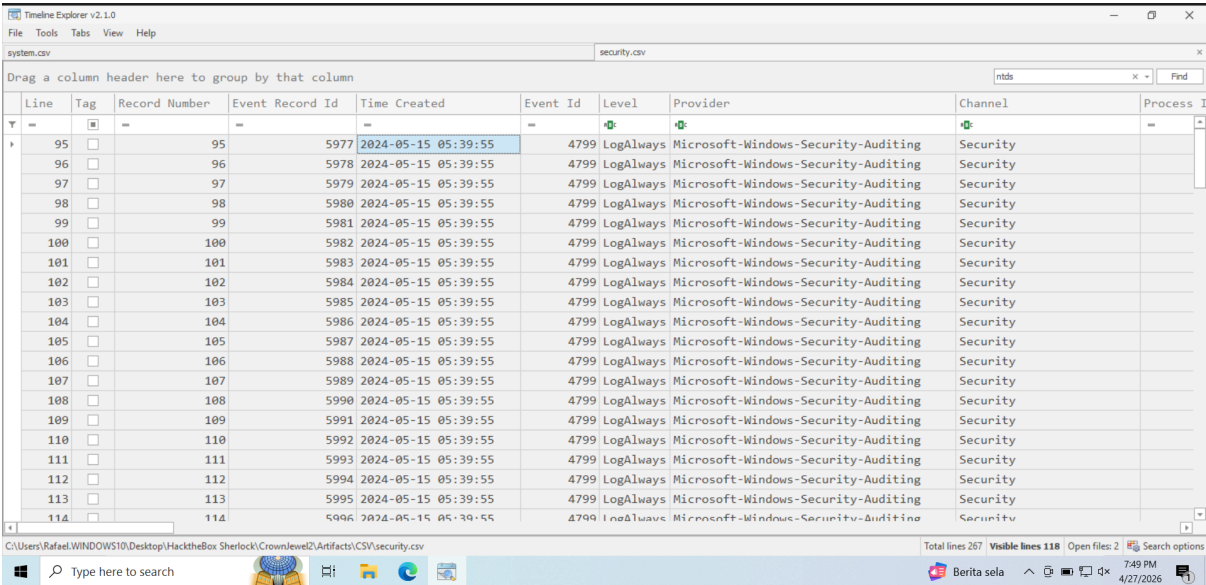
Processing C:\Users\Rafael.WINDOWS10\Desktop\HacktheBox Sherlock\CrownJewel2\Artifacts\APPLICATION.evtx...
Chunk count: 8, Iterating records...
Record #: 16 (timestamp: 2023-03-07 19:38:24.0439037): Warning! Time just went backwards! Last seen time before change:
2023-03-08 08:38:06.1082895

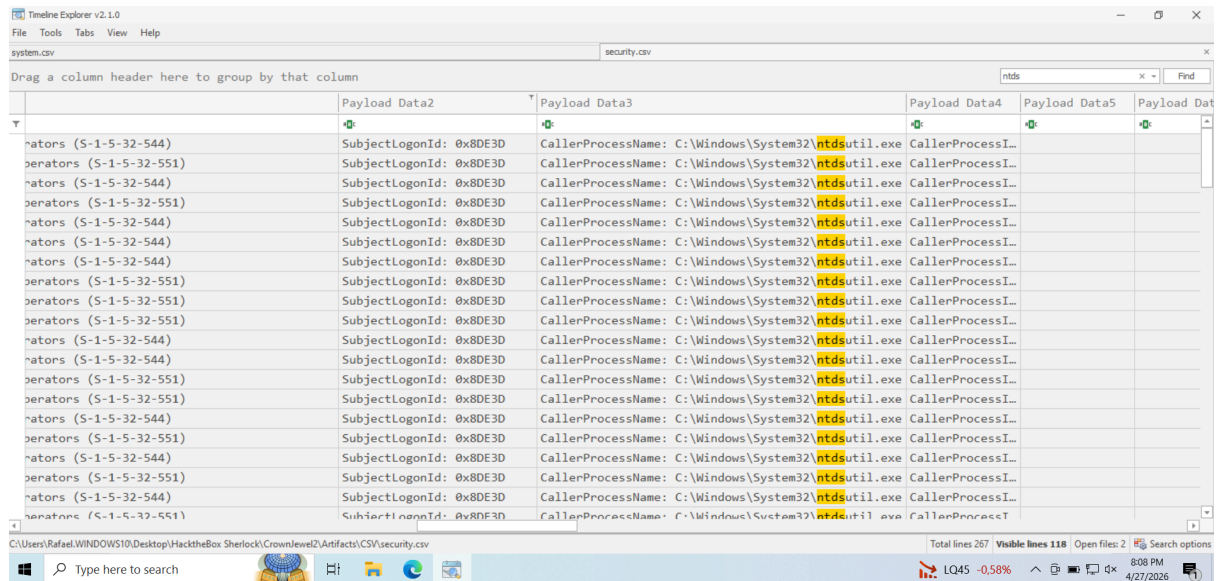
Event log details
Flags: None
Chunk count: 8
Stored/Calculated CRC: 1F304C9A/1F304C9A
Earliest timestamp: 2023-03-07 19:38:24.0439037
Latest timestamp: 2024-05-15 05:40:47.0960660
Total event log records found: 653

Records included: 653 Errors: 0 Events dropped: 0
```

Now that I have the CSV files, I can continue using Timeline Explorer to view the CSV in a nice GUI

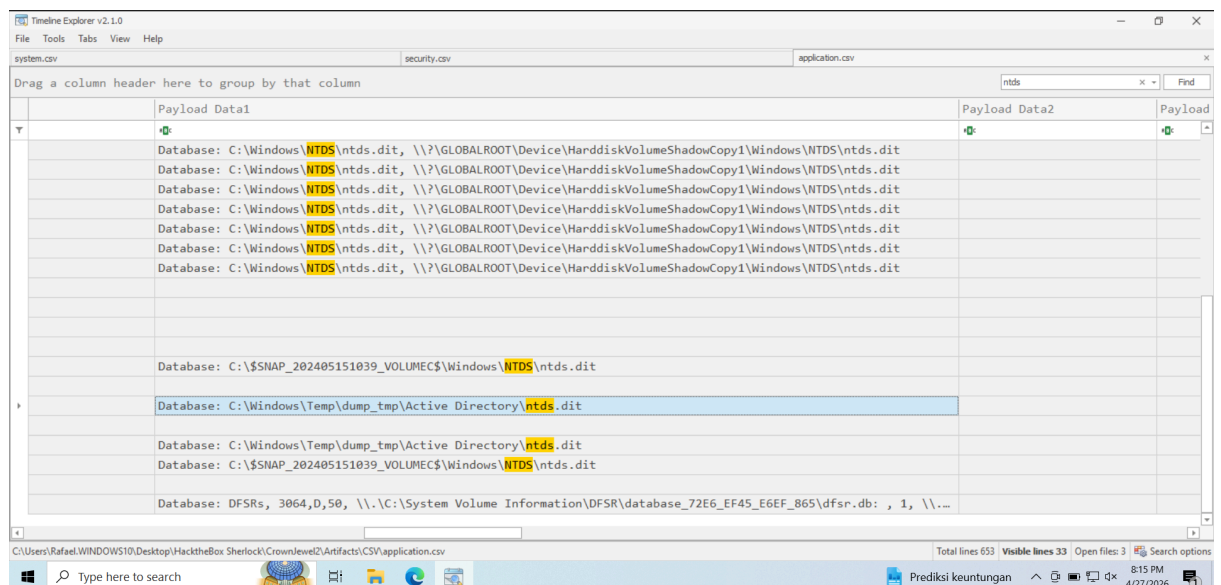
- 1. When utilizing ntdsutil.exe to dump NTDS on disk, it simultaneously employs the Microsoft Shadow Copy Service. What is the most recent timestamp at which this service entered the running state, signifying the possible initiation of the NTDS dumping process?





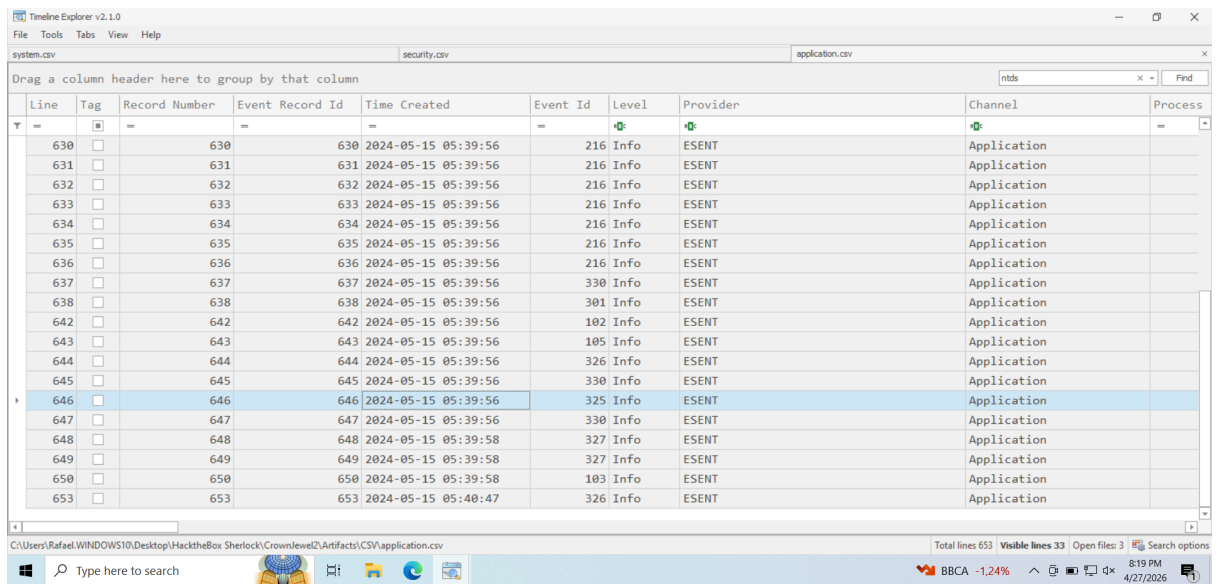
I filtered out the keyword "ntds" on security.csv. And then I found these results which is the ntdsutil.exe under the System32 folder, which was stated employed the Microsoft Shadow Copy Service, and I took the other the most recent time and that was the answer **2024-05-15 05:39:55**

2. Identify the full path of the dumped NTDS file.



Under the application.csv file I saw that there is a folder named dump_tmp containing the ntds.dit file which is a database for Active Directory which is also stored under the Temp folder. Therefore the answer is **C:\Windows\Temp\dump_tmp\Active Directory\ntds.dit**

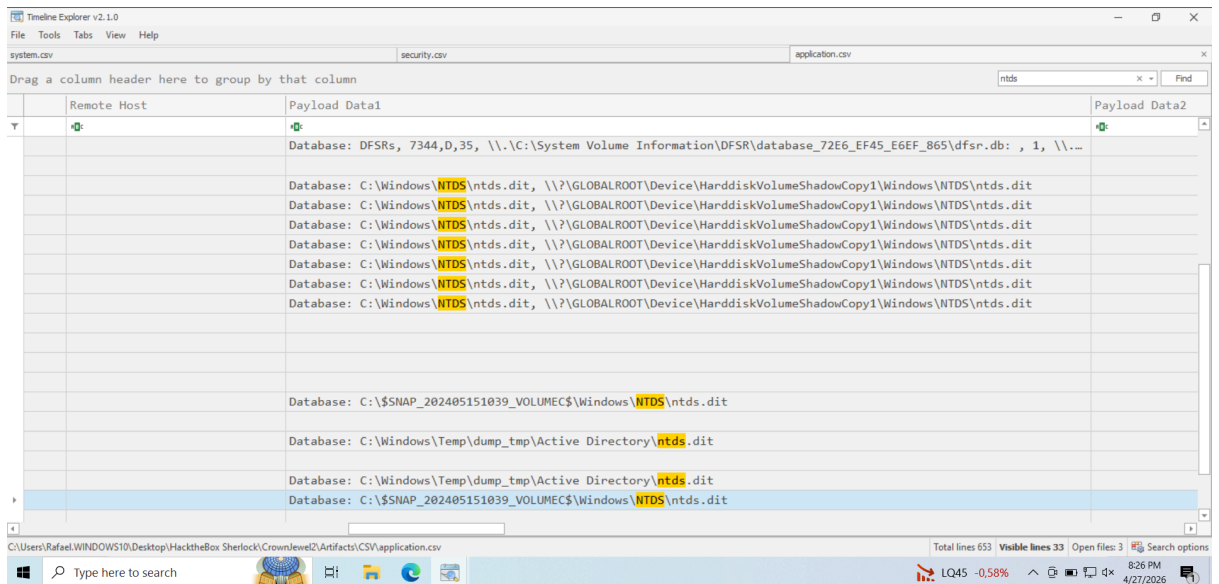
3. When was the database dump created on the disk?



Line	Tag	Record Number	Event Record Id	Time Created	Event Id	Level	Provider	Channel	Process
630		630	630	2024-05-15 05:39:56	216	Info	ESENT	Application	
631		631	631	2024-05-15 05:39:56	216	Info	ESENT	Application	
632		632	632	2024-05-15 05:39:56	216	Info	ESENT	Application	
633		633	633	2024-05-15 05:39:56	216	Info	ESENT	Application	
634		634	634	2024-05-15 05:39:56	216	Info	ESENT	Application	
635		635	635	2024-05-15 05:39:56	216	Info	ESENT	Application	
636		636	636	2024-05-15 05:39:56	216	Info	ESENT	Application	
637		637	637	2024-05-15 05:39:56	330	Info	ESENT	Application	
638		638	638	2024-05-15 05:39:56	301	Info	ESENT	Application	
642		642	642	2024-05-15 05:39:56	102	Info	ESENT	Application	
643		643	643	2024-05-15 05:39:56	105	Info	ESENT	Application	
644		644	644	2024-05-15 05:39:56	326	Info	ESENT	Application	
645		645	645	2024-05-15 05:39:56	330	Info	ESENT	Application	
646		646	646	2024-05-15 05:39:56	325	Info	ESENT	Application	
647		647	647	2024-05-15 05:39:56	330	Info	ESENT	Application	
648		648	648	2024-05-15 05:39:58	327	Info	ESENT	Application	
649		649	649	2024-05-15 05:39:58	327	Info	ESENT	Application	
650		650	650	2024-05-15 05:39:58	103	Info	ESENT	Application	
653		653	653	2024-05-15 05:40:47	326	Info	ESENT	Application	

Still under the exact same entry as the previous question, the time created is **2024-05-15 05:39:56**

4. When was the newly dumped database considered complete and ready for use?



Line	Remote Host	Payload Data1	Payload Data2
646		Database: DFSRs, 7344,D,35, \\.\C:\System Volume Information\DFSR\database_72E6_EF45_E6EF_865\dfsr.db: , 1, \\...	
647		Database: C:\Windows\NTDS\ntds.dit, \\?\GLOBALROOT\Device\HarddiskVolumeShadowCopy1\Windows\NTDS\ntds.dit	
648		Database: C:\Windows\NTDS\ntds.dit, \\?\GLOBALROOT\Device\HarddiskVolumeShadowCopy1\Windows\NTDS\ntds.dit	
649		Database: C:\Windows\NTDS\ntds.dit, \\?\GLOBALROOT\Device\HarddiskVolumeShadowCopy1\Windows\NTDS\ntds.dit	
650		Database: C:\Windows\NTDS\ntds.dit, \\?\GLOBALROOT\Device\HarddiskVolumeShadowCopy1\Windows\NTDS\ntds.dit	
651		Database: C:\Windows\NTDS\ntds.dit, \\?\GLOBALROOT\Device\HarddiskVolumeShadowCopy1\Windows\NTDS\ntds.dit	
652		Database: C:\Windows\NTDS\ntds.dit, \\?\GLOBALROOT\Device\HarddiskVolumeShadowCopy1\Windows\NTDS\ntds.dit	
653		Database: C:\Windows\NTDS\ntds.dit, \\?\GLOBALROOT\Device\HarddiskVolumeShadowCopy1\Windows\NTDS\ntds.dit	
654		Database: C:\\$SNAP_202405151039_VOLUMEC\$\Windows\NTDS\ntds.dit	
655		Database: C:\Windows\Temp\dump_tmp\Active Directory\ntds.dit	
656		Database: C:\Windows\Temp\dump_tmp\Active Directory\ntds.dit	
657		Database: C:\\$SNAP_202405151039_VOLUMEC\$\Windows\NTDS\ntds.dit	

Event Record Id	Time Created	Event Id	Level	Provider	Channel	Process Id	Computer	User Id
540	2024-05-14 04:01:50	325	Info	ESENT	Application	0	DC01.forela.lo...	
601	2024-05-15 05:35:48	1023	Error	Microsoft-Windows-PerFlib	Application	4612	DC01.forela.lo...	S-1-5-
630	2024-05-15 05:39:56	216	Info	ESENT	Application	0	DC01.forela.lo...	
631	2024-05-15 05:39:56	216	Info	ESENT	Application	0	DC01.forela.lo...	
632	2024-05-15 05:39:56	216	Info	ESENT	Application	0	DC01.forela.lo...	
633	2024-05-15 05:39:56	216	Info	ESENT	Application	0	DC01.forela.lo...	
634	2024-05-15 05:39:56	216	Info	ESENT	Application	0	DC01.forela.lo...	
635	2024-05-15 05:39:56	216	Info	ESENT	Application	0	DC01.forela.lo...	
636	2024-05-15 05:39:56	216	Info	ESENT	Application	0	DC01.forela.lo...	
637	2024-05-15 05:39:56	330	Info	ESENT	Application	0	DC01.forela.lo...	
638	2024-05-15 05:39:56	301	Info	ESENT	Application	0	DC01.forela.lo...	
642	2024-05-15 05:39:56	102	Info	ESENT	Application	0	DC01.forela.lo...	
643	2024-05-15 05:39:56	105	Info	ESENT	Application	0	DC01.forela.lo...	
644	2024-05-15 05:39:56	326	Info	ESENT	Application	0	DC01.forela.lo...	
645	2024-05-15 05:39:56	330	Info	ESENT	Application	0	DC01.forela.lo...	
646	2024-05-15 05:39:56	325	Info	ESENT	Application	0	DC01.forela.lo...	
647	2024-05-15 05:39:56	330	Info	ESENT	Application	0	DC01.forela.lo...	
648	2024-05-15 05:39:58	327	Info	ESENT	Application	0	DC01.forela.lo...	
649	2024-05-15 05:39:58	327	Info	ESENT	Application	0	DC01.forela.lo...	S-1-5-
650	2024-05-15 05:39:58	103	Info	FSFMT	Application	0	DC01.forela.lo...	

Below the answer to question number 3's answer, there seems to be another ntds.dit file which is located not under the system folder but instead looks like a legit folder, and I speculated that is the newly dumped database, and I got the time which is **2024-05-15 05:39:58**.

- Event logs use event sources to track events coming from different sources. Which event source provides database status data like creation and detachment?

Task 5

Look at the Event source in Events from question 2 to 4.

I originally did not understand how to do this question, therefore I opened the hint.

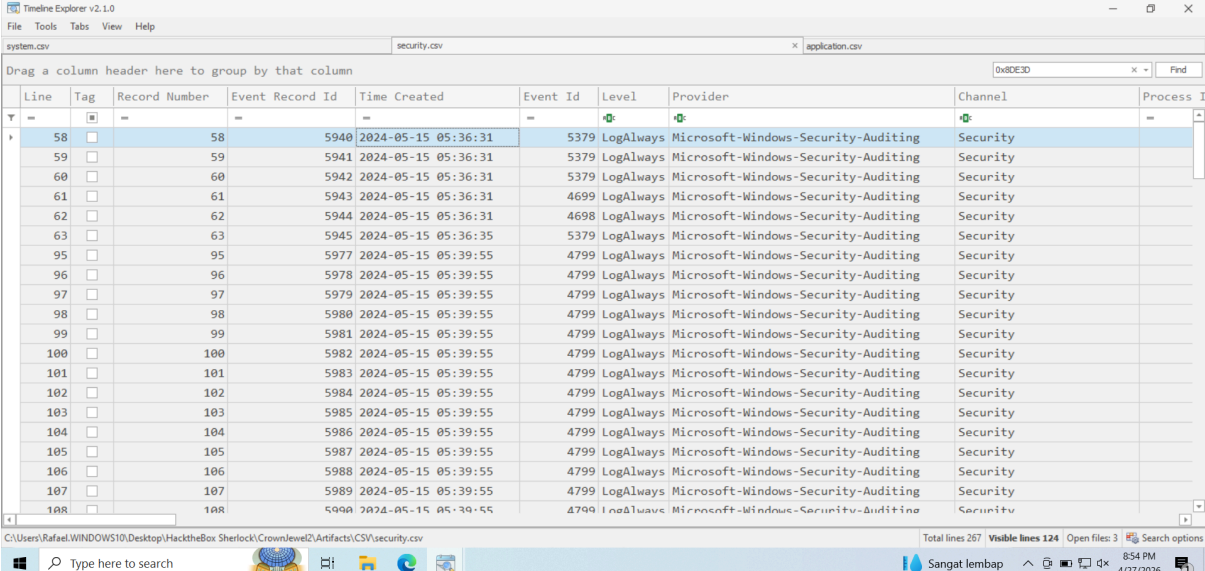
And then still in the same row as the previous number, in the screenshot above can be seen the provider or event source which is ESENT (Extensible Storage Engine Technology) is a built-in Windows transactional database engine, designed for high-performance, low-overhead storage of structured or semi-structured data

6. When `ntdsutil.exe` is used to dump the database, it enumerates certain user groups to validate the privileges of the account being used. Which two groups are enumerated by the `ntdsutil.exe` process? Give the groups in alphabetical order joined by comma space.

Here under the security.csv file there can be seen the ntdsutil.exe application like stated in the question but there also stated 2 user groups

to validate the privileges of the accounts. And I am tasked to order them alphabetically, therefore the answer is **Administrators, Backup Operators.**

7. Now you are tasked to find the Login Time for the malicious Session. Using the Logon ID, find the Time when the user logon session started.



Line	Tag	Record Number	Event Record Id	Time Created	Event Id	Level	Provider	Channel	Process I
58		58	5940	2024-05-15 05:36:31	5379	LogAlways	Microsoft-Windows-Security-Auditing	Security	
59		59	5941	2024-05-15 05:36:31	5379	LogAlways	Microsoft-Windows-Security-Auditing	Security	
60		60	5942	2024-05-15 05:36:31	5379	LogAlways	Microsoft-Windows-Security-Auditing	Security	
61		61	5943	2024-05-15 05:36:31	4699	LogAlways	Microsoft-Windows-Security-Auditing	Security	
62		62	5944	2024-05-15 05:36:31	4698	LogAlways	Microsoft-Windows-Security-Auditing	Security	
63		63	5945	2024-05-15 05:36:35	5379	LogAlways	Microsoft-Windows-Security-Auditing	Security	
95		95	5977	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	
96		96	5978	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	
97		97	5979	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	
98		98	5980	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	
99		99	5981	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	
100		100	5982	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	
101		101	5983	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	
102		102	5984	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	
103		103	5985	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	
104		104	5986	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	
105		105	5987	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	
106		106	5988	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	
107		107	5989	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	
108		108	5990	2024-05-15 05:39:55	4799	LogAlways	Microsoft-Windows-Security-Auditing	Security	

We can tell from the screenshot from the previous question that the LogonID is 0x8DE3D and I filtered it again in security.csv and I found several results, but I will take the most top which is when it started therefore the answer is **2024-05-15 05:36:31.**



You have solved CrownJewel-2!

Congratulations  **rafaelangelo** best of luck in capturing flags ahead!

#3195	27 Apr 2026	Retired
Sherlock Rank	Solve Date	Sherlock State

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